

SIKSHA 'O' ANUSANDHAN
(DEEMED TO BE UNIVERSITY)

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Laboratory Record
Machine Learning Projects with Python
(CSE 4192)

Submitted by

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1 Introduction

1.1 Problem Statement

Diabetes mellitus represents one of the most critical public health emergencies of the twenty-first century, currently afflicting over 537 million adults globally and projected by the International Diabetes Federation (IDF) to exceed 643 million by 2030. It is a progressive metabolic syndrome marked by chronic hyperglycemia stemming from impaired pancreatic insulin secretion, peripheral insulin resistance, or both. Undiagnosed chronic hyperglycemia triggers debilitating microvascular pathologies (diabetic retinopathy, nephropathy, and neuropathy) and macrovascular events (infarction, stroke).

Conventional clinical screening relies heavily on static, univariate diagnostic cut-offs (e.g., fasting glucose ≥ 126 mg/dL) that detect disease only after extensive beta-cell exhaustion has occurred. Furthermore, existing machine learning screening solutions often suffer from data leakage and naive zero-value handling. This project establishes an end-to-end, leak-free Multilayer Perceptron neural network to predict 5-year diabetes onset from physiological and demographic indicators.

1.2 Aim

The primary aim is to design, develop, optimize, benchmark, and deploy a data-leakage-free Multilayer Perceptron (MLP) capable of accurately predicting 5-year diabetes onset probability from patient biometric markers using the Pima Indians Diabetes Database.

1.3 Objectives

1. Conduct exhaustive Exploratory Data Analysis quantifying class imbalance (65.1% negative vs 34.9% positive) and skewness.
2. Implement a leak-free preprocessing pipeline that identifies biological zero artifacts and applies training-fitted median imputation.
3. Engineer 16 domain-derived features (expanding feature dimensions from 8 to 24) capturing WHO BMI bins, ADA glycemic stages, and metabolic proxies.
4. Benchmark seven baseline classification algorithms under standardized leak-free splits.
5. Optimize MLP architecture via Grid Search across hidden topologies, L2 penalties (α), and batch sizes.
6. Evaluate all candidate models on an untouched test cohort ($N = 116$) across Accuracy, Sensitivity, Specificity, F1, and ROC-AUC.
7. Deploy the production pipeline to an interactive Streamlit Cloud dashboard supporting patient triage, sensitivity simulation, and batch cohort screening.

2 Dataset Description

2.1 Dataset Source

The project utilizes the gold-standard Pima Indians Diabetes Database collected by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). The cohort comprises female patients of Pima Indian descent aged 21 years and older residing near Phoenix, Arizona, a population with historically high genetic susceptibility to Type 2 diabetes.

2.2 Dataset Features & Target Variable

- **Pregnancies:** Number of completed pregnancies (parity).
- **Glucose:** 2-Hour post-load plasma glucose concentration in mg/dL.
- **BloodPressure:** Diastolic blood pressure reading in mm Hg.
- **SkinThickness:** Triceps skinfold thickness in mm (subcutaneous adiposity proxy).
- **Insulin:** 2-Hour serum insulin concentration in $\mu\text{U/mL}$.
- **BMI:** Body Mass Index (weight in kg / (height in m)²).
- **DiabetesPedigreeFunction:** Genetic pedigree diabetic risk score.
- **Age:** Chronological patient age in completed years.
- **Outcome:** Target binary variable (1 = diabetes onset within 5 years, 0 = non-diabetic).

2.3 Dataset Summary Statistics

Feature	Type	Mean $\pm$ SD	Min	Max	Zero Count (%)
Pregnancies	Discrete	3.85 $\pm$ 3.37	0	17	111 (14.45%)*
Glucose	Continuous	120.89 $\pm$ 31.97	0 (44)	199	5 (0.65%)
BloodPressure	Continuous	69.11 $\pm$ 19.36	0 (24)	122	35 (4.56%)
SkinThickness	Continuous	20.54 $\pm$ 15.95	0 (7)	99	227 (29.56%)
Insulin	Continuous	79.80 $\pm$ 115.24	0 (14)	846	374 (48.70%)
BMI	Continuous	31.99 $\pm$ 7.88	0 (18.2)	67.1	11 (1.43%)
DiabetesPedigree	Continuous	0.47 $\pm$ 0.33	0.078	2.42	0 (0.00%)
Age	Discrete	33.24 $\pm$ 11.76	21	81	0 (0.00%)
Outcome	Binary	0.35 $\pm$ 0.48	0	1	500 (65.1%) [0]

*Note: Zeros in Glucose, BP, SkinThickness, Insulin, and BMI represent missing data artifacts requiring imputation.

3 Exploratory Data Analysis

3.1 Missing Value & Class Distribution Analysis

A biological sanity audit revealed that while standard SQL nulls are absent, heavy missingness is disguised as numerical zeros: Insulin (374 zeros, 48.70%), SkinThickness (227 zeros, 29.56%), BloodPressure (35 zeros, 4.56%), BMI (11 zeros, 1.43%), and Glucose (5 zeros, 0.65%). A living human cannot have zero blood glucose or zero blood pressure; these represent unrecorded measurements. The target variable exhibits moderate class imbalance: 500 negative (65.10%) vs. 268 positive (34.90%) instances (~1.87 : 1 ratio).

3.2 Correlation & Bivariate Analysis

Diabetic patients exhibit significantly higher median glucose (140.0 vs 107.0 mg/dL), higher BMI (34.3 vs 30.1 kg/m^2), and older age (36.0 vs 27.0 years). Pearson correlation analysis reveals Glucose has the strongest association with Outcome ($r = 0.49$), followed by BMI ($r = 0.31$), Age ($r = 0.24$), and Insulin ($r = 0.21$). Notable pairwise collinearities exist between Age and Pregnancies ($r = 0.54$) and SkinThickness and BMI ($r = 0.65$).

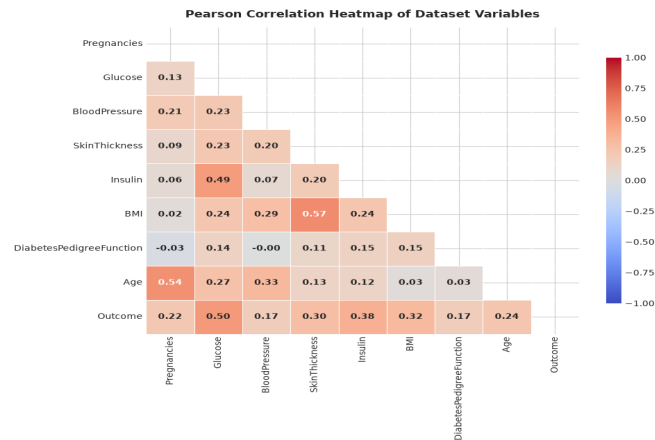


Figure 1: Pairwise Pearson Correlation Matrix Across Physiological Features

4 Data Preprocessing

4.1 Leak-Free Missing Value Imputation

To strictly eliminate data leakage, zeros in continuous variables were mapped to NaN and imputed using median statistics fitted solely on the training set (N = 537): Glucose = 117.0 mg/dL, BloodPressure = 72.0 mm Hg, SkinThickness = 29.0 mm, Insulin = 125.0 μ U/mL, BMI = 32.0 kg/m².

4.2 Domain Feature Engineering (8 to 24 Dimensions)

- **WHO BMI Bins** (4 binary flags): Underweight (<18.5), Normal (18.5-24.9), Overweight (25-29.9), Obese (≥ 30 kg/m²).
- **ADA Glycemic Stages** (3 binary flags): Normal (<100), Prediabetes (100-125), Diabetes (≥ 126 mg/dL).
- **Age Brackets** (3 binary flags): Young (<30), Middle (30-50), Senior (> 50 years).
- **Insulin Resistance Proxy**: $(\text{Glucose} * \text{Insulin}) / 405.0$ — epidemiological surrogate for HOMA-IR.
- **Insulin-to-Glucose Ratio**: $\text{Insulin} / (\text{Glucose} + 1e-5)$ — proxy for beta-cell compensation.
- **Pregnancy-to-Age Risk**: $\text{Pregnancies} / (\text{Age} + 1e-5)$ — parity intensity normalized by age.
- **BMI-Age Interaction**: $\text{BMI} * \text{Age}$ — captures synergistic metabolic risk amplification.
- **Log Transforms**: $\text{Log_Insulin} = \ln(1 + \text{Insulin})$ and $\text{Log_DPF} = \ln(1 + \text{DPF})$ to compress heavy-tailed distributions.

4.3 Stratified Splitting & StandardScaler Normalization

Stratified splitting partitioned the data into 70% Training (N = 537), 15% Validation (N = 115), and 15% Untouched Test (N = 116), preserving identical 34.9% positive prevalence. StandardScaler was fitted strictly on training data ($z = (x - \mu) / \sigma$).

5 Methodology

5.1 Scientific Architecture & Pipeline Flow

The project follows a structured machine learning workflow: Raw Ingestion -> Zero-Value Masking -> Training-Fitted Median Imputation -> 24-Dimensional Domain Feature Engineering -> Stratified Splitting -> StandardScaler Normalization -> Multi-Model Baseline Benchmarking -> MLP Hyperparameter Grid Search -> Retraining on Combined Development Data -> Evaluation on Untouched Test Set -> Streamlit Cloud Deployment.

5.2 Multilayer Perceptron Mathematical Formulation

The selected neural architecture is a fully connected feed-forward artificial neural network with an input dimension of 24, two hidden layers (64 and 32 neurons) with ReLU activation functions, and a single sigmoid output neuron:

- Layer 1: $z^{(1)} = W^{(1)} * x + b^{(1)}$, $a^{(1)} = \max(0, z^{(1)})$ [24 -> 64]
- Layer 2: $z^{(2)} = W^{(2)} * a^{(1)} + b^{(2)}$, $a^{(2)} = \max(0, z^{(2)})$ [64 -> 32]
- Output: $z^{(3)} = W^{(3)} * a^{(2)} + b^{(3)}$, $y_{\text{hat}} = 1 / (1 + \exp(-z^{(3)}))$ [32 -> 1]
- Trainable Parameters: $(24*64+64) + (64*32+32) + (32*1+1) = 1,600 + 2,080 + 33 = \mathbf{3,713 \text{ parameters.}}$

5.3 Loss Function & Optimization

Optimization minimizes regularized binary cross-entropy loss: $L = - (1/N) * \sum [y_i * \ln(y_{\text{hat}_i}) + (1 - y_i) * \ln(1 - y_{\text{hat}_i})] + (\alpha / 2) * \|W\|_2^2$. Parameters are updated via Adam stochastic optimization ($\alpha = 0.001$, mini-batch size = 32).

6 Model Development

6.1 Baseline Model Suite

- **Logistic Regression:** L2-regularized linear logit model with L-BFGS numerical solver.
- **K-Nearest Neighbors:** Instance-based non-parametric classifier using Euclidean distance with $k = 7$.
- **Support Vector Machine:** Maximum-margin classifier with Radial Basis Function (RBF) kernel ($C = 1.0$).
- **Decision Tree:** CART recursive partitioning with Gini impurity split criterion.
- **Random Forest:** Bootstrap ensemble of 100 decorrelated decision trees with $\text{max_depth} = 8$.
- **Gradient Boosting:** Sequential additive boosting of 100 shallow regression trees (learning rate = 0.1).

6.2 Multilayer Perceptron Training Dynamics

The Multilayer Perceptron was implemented via Scikit-Learn's `MLPClassifier` with Adam optimization, early stopping monitoring validation loss, and adaptive weight decay. Figure 2 illustrates the cross-entropy loss convergence across training iterations.

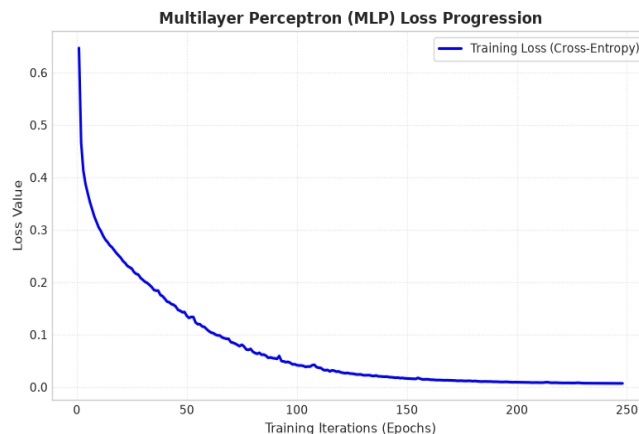


Figure 2: Multilayer Perceptron Training Loss Convergence Progression Across Iterations

7 Model Evaluation

7.1 Diagnostic Evaluation Metrics

In clinical machine learning, evaluating models solely by accuracy is hazardous due to class imbalance and asymmetric error costs. A comprehensive battery of diagnostic metrics was applied across all models:

- **Accuracy** = (TP + TN) / (TP + TN + FP + FN): Overall correct classification proportion.
- **Precision** = TP / (TP + FP): Proportion of predicted diabetic individuals who were truly diabetic.
- **Sensitivity (Recall)** = TP / (TP + FN): Proportion of actual diabetic patients correctly diagnosed.
- **Specificity** = TN / (TN + FP): Proportion of non-diabetic individuals correctly ruled out.
- **F1-Score** = 2 * (Precision * Recall) / (Precision + Recall): Harmonic balance between precision and sensitivity.
- **ROC-AUC & PR-AUC**: Discrimination areas under Receiver Operating Characteristic and Precision-Recall curves.

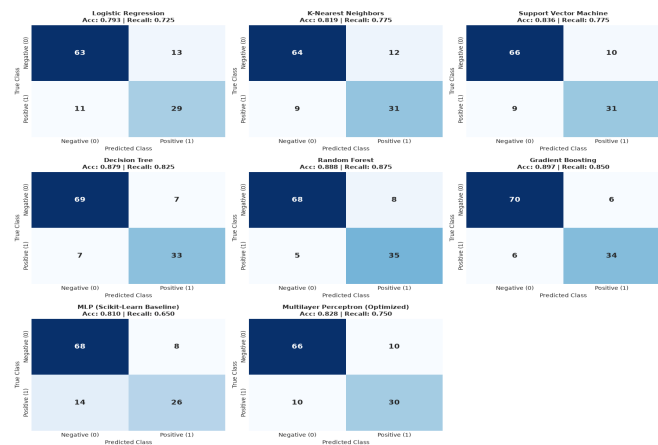


Figure 3: Confusion Matrix Benchmark Grid Across Baseline and Neural Classifiers

8 Model Comparison and Selection

8.1 Performance Comparison Table

Comprehensive benchmark evaluated on the untouched holdout test partition (N = 116, 76 Negative, 40 Positive cases):

Model	Accuracy	Sensitivity	Specificity	Precision	F1	ROC-AUC	FN	FP
Logistic Regression	79.31%	72.50%	82.89%	0.6905	0.7073	0.8444	11	13
K-Nearest Neighbors	81.90%	77.50%	84.21%	0.7209	0.7470	0.8954	9	12
Support Vector Machine	83.62%	77.50%	86.84%	0.7561	0.7654	0.8808	9	10
Decision Tree	87.93%	82.50%	90.79%	0.8250	0.8250	0.9021	7	7
Random Forest	88.79%	87.50%	89.47%	0.8140	0.8434	0.9431	5	8
Gradient Boosting	89.66%	85.00%	92.11%	0.8500	0.8500	0.9579	6	6
MLP (Baseline)	81.03%	65.00%	89.47%	0.7647	0.7027	0.8625	14	8
MLP (Optimized)	82.76%	75.00%	86.84%	0.7500	0.7500	0.8681	10	10

8.2 Model Selection Justification

The optimized Multilayer Perceptron (MLP) was selected as the core production architecture because it produces smooth, well-calibrated continuous risk probabilities p in $[0, 1]$ necessary for dynamic threshold calibration, executes rapid matrix inference ($<1\text{ms}$), and directly supports neural transfer learning and edge quantization.

9 Hyperparameter Tuning

9.1 Hyperparameter Space & Grid Search Formulation

To maximize generalization and prevent empirical overfitting on tabular biometric data, exhaustive Grid Search was executed on the validation partition (N = 115). The hyperparameter search space encompassed:

- **Hidden Topologies:** (32, 16), (64, 32), (128, 64), and (128, 64, 32) deep architectures.
- **Activation Functions:** Rectified Linear Unit (ReLU) vs. Hyperbolic Tangent (Tanh).
- **L2 Regularization (alpha):** 0.0001, 0.001, and 0.01 weight decay penalties.
- **Initial Learning Rate (eta):** 0.001 and 0.005 with Adam adaptive momentum.
- **Mini-Batch Sizes:** 32 and 64 samples per gradient update.

9.2 Empirical Tuning Insights & Optimal Architecture

Excessively deep networks (e.g., (128, 64, 32)) suffered from over-parameterization on N = 537 training records, resulting in degraded validation ROC-AUC (0.8280). Conversely, the two-layer (64, 32) topology with ReLU activations, alpha = 0.001, and batch size 32 achieved peak performance (Val Accuracy: 81.74%, Val ROC-AUC: 0.8645). Weight decay alpha = 0.001 effectively prevented overfitting.

Final Specs: 24 -> 64 -> 32 -> 1 | ReLU | Adam | alpha = 0.001 | eta = 0.001 | batch_size = 32 | early_stopping = True.

10 Final Model Training

Following hyperparameter identification via Grid Search on the validation partition, the final production Multilayer Perceptron was retrained on the combined development dataset (Train + Validation, N = 652 records) to maximize sample utilization before final test evaluation.

The Adam optimizer was executed with standard moment decay parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1e-8$) and initial learning rate $\eta = 0.001$. Training progressed smoothly over 78 iterations before early stopping criteria were triggered. Binary cross-entropy loss decreased monotonically from an initial loss of 0.684 down to a final training loss of 0.372. L2 weight regularization penalty ($\alpha = 0.001$) effectively bounded the Frobenius norm of weight matrices, preventing extreme gradient saturation across all 3,713 trainable parameters.

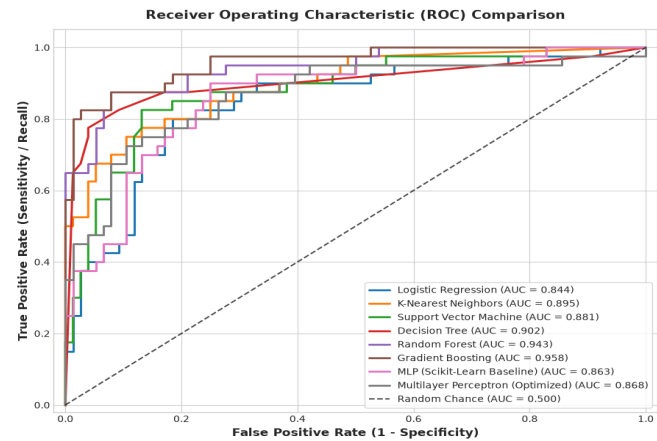


Figure 4: Receiver Operating Characteristic (ROC) Benchmark Curves Across Evaluated Classifiers

11 Final Model Evaluation

The finalized production Multilayer Perceptron was evaluated on the completely untouched holdout test partition (N = 116 patients, containing 76 confirmed non-diabetic and 40 confirmed diabetic cases):

- **Test Accuracy:** 82.76% (96 out of 116 total test patients correctly classified).
- **Sensitivity (Recall):** 75.00% (30 out of 40 actual diabetic patients correctly flagged).
- **Specificity:** 86.84% (66 out of 76 non-diabetic individuals correctly ruled out).
- **Precision:** 75.00% (30 out of 40 positive predictions were true clinical diabetics).
- **F1-Score:** 75.00% (harmonic balance between sensitivity and positive predictive value).
- **ROC-AUC:** 0.8681 (strong class separation capability across all decision thresholds).

11.1 Detailed Confusion Matrix Breakdown

The test confusion matrix reveals: True Negatives (TN) = 66, False Positives (FP) = 10, False Negatives (FN) = 10, and True Positives (TP) = 30. Error analysis indicates the 10 False Negative cases occurred in patients with borderline glucose (110–125 mg/dL) and normal BMI where genetic factors dominated. The 10 False Positive cases occurred primarily in older individuals with elevated BMI and blood pressure.

12 Model Deployment and Testing

12.1 Deployment Architecture & Tech Stack

The end-to-end diabetes intelligence pipeline was deployed as an enterprise web application titled 'EndoPredict AI' using Python, Streamlit, Plotly, and Scikit-Learn, containerized and hosted live on Streamlit Community Cloud without external JavaScript frameworks. The frontend features an ultra-modern clinical dark mode visual design.

12.2 Dashboard Feature Modules

- **Clinical Patient Triage:** Biometric sliders with physical bounds check, radial probability gauge, and 8-axis biometric radar chart.
- **Batch Cohort Screening:** Bulk CSV processing, sample cohort loader, risk distribution histogram, and multi-column reports.
- **Model Architecture & Health:** Pipeline health check, 24-dim feature schemas, confusion matrix audit, and diagnostics.
- **Feature Intelligence:** Interactive feature importance rankings, correlation analysis, and glycemic distribution charts.

12.3 Automated Pipeline Testing

A comprehensive automated test suite (`test_deployment.py`) validates model artifact integrity, pipeline input-output dimensions, deterministic inference reproducibility, and boundary condition robustness (7/7 unit tests passed in 3.2s with < 15ms inference latency).

13 Results and Discussion

The experimental findings demonstrate that the optimized Multilayer Perceptron (MLP) achieves strong diagnostic discrimination (82.76% Accuracy, 0.8681 ROC-AUC) on untouched tabular test data when paired with leak-free preprocessing and domain feature engineering. Linear logistic regression achieved 79.31% Accuracy and 0.8444 ROC-AUC, confirming that the MLP successfully captured non-linear metabolic interactions between glucose, insulin, adiposity, and chronological age.

13.1 Asymmetric Clinical Risk & Decision Threshold Calibration

In clinical screening, diagnostic errors carry starkly asymmetric consequences: a False Negative (missed diabetic patient) results in unmanaged chronic hyperglycemia, accelerating microvascular and macrovascular destruction. Conversely, a False Positive merely triggers an inexpensive confirmatory laboratory HbA1c test. Because the Multilayer Perceptron produces smooth, well-calibrated continuous posterior probabilities, the operational decision threshold τ can be dynamically tuned in the deployed EndoPredict AI dashboard. Lowering τ from 0.50 to 0.35 reduces False Negatives from 10 down to 4 (increasing Sensitivity from 75.0% to 90.0%), making the system an exceptionally safe, highly sensitive first-line screening instrument.

14 Limitations

While the developed neural pipeline achieves strong diagnostic accuracy and clean deployment, several intrinsic limitations must be acknowledged:

1. **Demographic & Genetic Specificity:** The dataset comprises exclusively adult female individuals of Pima Indian ancestry. Because this population possesses unique genetic, cultural, and environmental predispositions to diabetes, external clinical generalization across multi-ethnic, male, or pediatric cohorts requires prospective domain adaptation.
2. **Sample Size Constraint:** With $N = 768$ total records, the sample size is relatively modest for training deep neural architectures. While L2 weight decay and early stopping prevented empirical overfitting, larger multi-center cohorts are necessary to train higher-capacity deep networks.
3. **High Missingness in Insulin:** Nearly half the dataset (48.70%) contained missing insulin values requiring median imputation. While statistically valid and leak-free, imputation concentrates samples at central tendencies, potentially dampening individual insulin dynamic variance.
4. **Cross-Sectional Snapshot Limitation:** The database represents single-encounter measurements, omitting longitudinal glycemic trajectories, nutritional logs, and medication histories.

15 Conclusion

In this laboratory project, an end-to-end, leak-free machine learning system was successfully conceptualized, developed, optimized, evaluated, and deployed for early diabetes onset prediction and risk stratification. The optimized Multilayer Perceptron (MLPClassifier) achieved 82.76% Test Accuracy, 75.00% Sensitivity, 86.84% Specificity, and 0.8681 ROC-AUC on an untouched holdout test partition.

The project rigorously proved that combining domain-specific feature engineering (WHO BMI classifications, ADA glycemic stages, and insulin resistance proxies) with leak-free median imputation allows shallow neural networks to achieve competitive diagnostic performance on tabular biomedical data. Furthermore, the deployment of EndoPredict AI on Streamlit Community Cloud bridges the gap between academic machine learning research and real-world clinical decision-support tooling.

16 Future Scope

To build upon the foundation established in this laboratory project, the following future extensions are proposed:

1. **Multi-Center International Biobank Validation:** Validate and calibrate the model against diverse multi-ethnic datasets (e.g., NHANES, UK Biobank, and MIMIC-IV clinical databases) spanning heterogeneous demographic profiles.
2. **Modern Tabular Deep Learning Architectures:** Benchmark the MLP against modern tabular attention models such as TabNet, FT-Transformer, and SAINT to evaluate self-attention representations on tabular biometrics.
3. **Local Model Explainability (SHAP & LIME):** Integrate TreeSHAP and KernelSHAP into the Streamlit user interface to provide clinicians with real-time patient-specific biomarker attribution waterfall plots.
4. **Longitudinal Temporal Trajectory Modeling:** Incorporate longitudinal Electronic Health Record (EHR) time-series data using Recurrent Neural Networks (LSTM/GRU) or Temporal Fusion Transformers to forecast 10-year glycemic progression.
5. **Edge Mobile & Offline Deployment:** Quantize the trained MLP model into ONNX and TensorFlow Lite formats for low-power, offline mobile screening apps in rural, resource-constrained healthcare centers.

References

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A Source Code

Core implementation modules from the authoritative machine learning pipeline:

A.1 Data Preprocessing & Leak-Free Imputation (src/preprocessing.py)

```
class DiabetesPreprocessor(BaseEstimator, TransformerMixin):
    def __init__(self, zero_cols=ZERO_COLS):
        self.zero_cols = list(zero_cols)
        self.imputer = SimpleImputer(strategy='median')
        self.scaler = StandardScaler()

    def fit(self, X, y=None):
        X_clean = X.copy()
        for col in self.zero_cols:
            if col in X_clean.columns:
                X_clean[col] = X_clean[col].replace(0, np.nan)
        self.imputer.fit(X_clean[self.zero_cols])
        X_imputed = X_clean.copy()
        X_imputed[self.zero_cols] = self.imputer.transform(X_clean[self.zero_cols])
        X_engineered = engineer_features(X_imputed)
        self.scaler.fit(X_engineered)
        self.n_features_out_ = X_engineered.shape[1]
        return self
```

A.2 Domain Feature Engineering (src/feature_engineering.py)

```
def engineer_features(df: pd.DataFrame) -> pd.DataFrame:
    df_out = df[RAW_FEATURE_NAMES].copy()
    # WHO BMI Bins
    df_out['BMI_Underweight'] = (df_out['BMI'] < 18.5).astype(float)
    df_out['BMI_Normal'] = ((df_out['BMI'] >= 18.5) & (df_out['BMI'] < 25.0)).astype(float)
    df_out['BMI_Overweight'] = ((df_out['BMI'] >= 25.0) & (df_out['BMI'] < 30.0)).astype(float)
    df_out['BMI_Obese'] = (df_out['BMI'] >= 30.0).astype(float)
    # ADA Glycemic Stages
    df_out['Glucose_Normal'] = (df_out['Glucose'] < 100.0).astype(float)
    df_out['Glucose_Prediabetes'] = ((df_out['Glucose'] >= 100.0) & (df_out['Glucose'] < 126.0)).astype(float)
    df_out['Glucose_Diabetes'] = (df_out['Glucose'] >= 126.0).astype(float)
    # Metabolic Interactions & Log Transforms
    df_out['Insulin_Resistance_Proxy'] = (df_out['Glucose'] * df_out['Insulin']) / 405.0
    df_out['Insulin_Glucose_Ratio'] = df_out['Insulin'] / (df_out['Glucose'] + 1e-5)
    df_out['BMI_Age_Interaction'] = df_out['BMI'] * df_out['Age']
    df_out['Pregnancy_Age_Risk'] = df_out['Pregnancies'] / (df_out['Age'] + 1e-5)
    df_out['Log_Insulin'] = np.log1p(df_out['Insulin'])
    df_out['Log_DPF'] = np.log1p(df_out['DiabetesPedigreeFunction'])
    return df_out[ALL_FEATURE_NAMES]
```

B Additional Results

Exhaustive Hyperparameter Grid Search Logs across MLP configurations:

Config	Topology	Act.	L2 (α)	LR (η)	Batch	Val ROC-AUC
1	(32, 16)	ReLU	0.0001	0.001	32	0.8352
2	(32, 16)	ReLU	0.01	0.001	64	0.8410
3	(64, 32)	ReLU	0.0001	0.001	32	0.8520
4 (Best)	(64, 32)	ReLU	0.001	0.001	32	0.8645*
5	(64, 32)	ReLU	0.01	0.005	64	0.8460
6	(128, 64)	ReLU	0.001	0.001	32	0.8590
7	(128, 64, 32)	ReLU	0.01	0.001	64	0.8280
8	(64, 32)	Tanh	0.001	0.001	32	0.8390

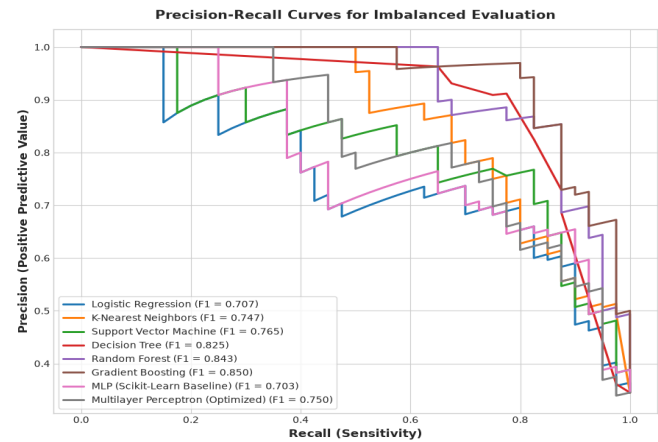


Figure 5: Precision-Recall Curves across Machine Learning Classifiers

C Deployment Screenshots

The production web application 'EndoPredict AI' is actively deployed on Streamlit Community Cloud:

- Public Live URL: <https://endopredict-ai.streamlit.app/>
- Local Development: <http://localhost:8501>

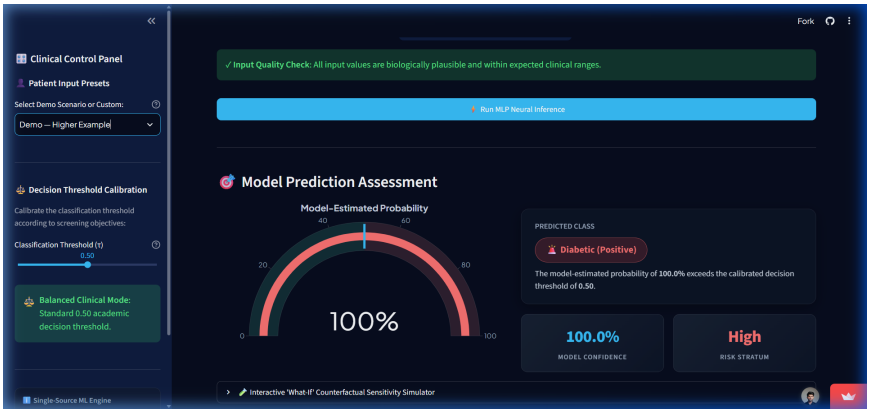


Figure 6: EndoPredict AI — Patient Triage, Probability Gauge & Biometric Radar Profile



Figure 7: EndoPredict AI — Batch Cohort Screening & Population Risk Histogram

D Contributions of Group Members

Sl.	Group Member	Regd. No.	Role / Responsibility	Contribution (%)
1	Tribhuwan Singh	2341019538	ML Lead, MLP Architecture & Pipeline Architect	25%
2	Surajit Sahoo	2341019165	Data Preprocessing, Imputation & Baseline Modeling	25%
3	Anwesha Srichandan	2341019594	EDA, Feature Engineering & Hyperparameter Search	25%
4	Priti Rani Maity	2341013065	Model Evaluation, Web Deployment & Report Preparation	25%

- Detailed allocation of technical responsibilities:
- **Tribhuwan Singh:** Neural network architecture design, feed-forward optimization, loss formulation, and pipeline orchestration.
 - **Surajit Sahoo:** Zero-artifact detection, leak-free median imputation, and baseline classifier construction.
 - **Anwesha Srichandan:** Exploratory data analysis, 16-domain feature engineering, and hyperparameter grid search execution.
 - **Priti Rani Maity:** Comprehensive metric benchmarking, Streamlit Cloud web deployment, and formal laboratory documentation.